Shelter Beacon

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A wall box that lets a disaster shelter ask for help with no phones, no internet and no grid power.

What it does

A shelter leader flips a switch for each thing the shelter needs and turns a knob to roughly how many people are there. The box radios that straight to headquarters, which ranks every shelter by urgency and rings an alarm for the worst. Headquarters confirms each report, and the box beeps twice so the leader knows it arrived.

Why

After the 2024 Noto Peninsula earthquake, 49 locations were cut off, most mobile coverage in the hardest-hit towns was lost, and fewer than half of the cut-off places had been on the prefecture's list. Japan's two-way disaster radio is built for staffed sites. Small informal shelters had no way to report.

The report

`PKT,42,W1,M0,FD0,SH0,FU0,O20,H0`

31 bytes, the whole state every time. Sent up to three times until headquarters confirms.

Priority at headquarters

Medical 3, water 2, food, blankets and fuel 1 each. 25 or more people add 1, 50 or more add 2. Late check-ins add 1, lost contact 2. A score of 3 or more is high priority.

Specification

Inputs	5 toggle switches: water, medical, food, blankets, fuel. An 11-position people knob, 0 to 100+
Feedback	Amber status light, beeper
Board	Heltec V4.3: ESP32-S3, SX1262
Radio	LoRa, 915 MHz, SF9, 125 kHz, CR 4/7
Time on air	≈ 0.31 s per copy
Weakest decoded	−122 dBm in range test 1
Power	4 × 18650, ≈ 52 Wh; 1 W solar panel
Runtime, no sun	≈ 8 days, estimated
Enclosure	290 × 190 × 140 mm hinged ABS box
Headquarters	Heltec V3 receiver, one-file dashboard over USB
Parts cost	≈ $107 for the first box

Status

Working class prototype, ENGR 1113, Fall 2026. Shown at one expo. Confirm-and-retry firmware not yet tested on hardware. 915 MHz is a US band: use in Japan needs a certified 920 MHz radio. Not certified for emergency use.

Try it, read the research, get the files
cale-star.github.io/shelter-beacon

Cale French, 2026. Code and files: github.com/Cale-star/shelter-beacon